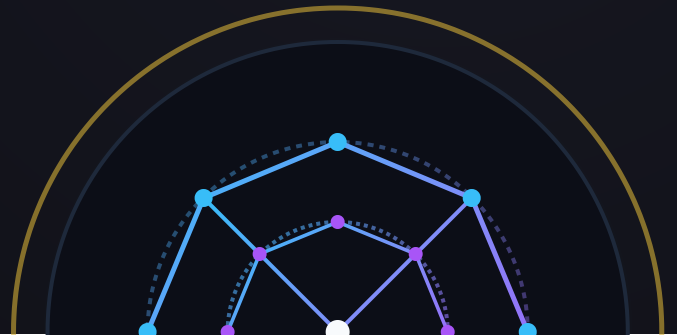
CODE KERNEL '26 · DECODE ROUND

NEUROVA

360° Multimodal Cognitive Learning & Diagnostic Engine
For Indian Classrooms (Classes 6 To 10).



Team Information & Institutional Details

Team Info Field	Official Submission Details
Team Leader Name	Anurag
Team Leader Mobile Number	8247078530
Team Leader Email	varmaanurag2008@gmail.com
No. of Team Members	4
Team Members Details	Member 1: Anurag | Member 2: Siddarth | Member 3: Solmon | Member 4: Jithin
College Name	Gokaraju Rangaraju Institute of Engineering and Technology (GRIET)

★ NEUROVA MISSION: Built at GRIET to solve the Indian classroom pacing crisis and put school teachers back in command.

Why One Pace Fails Indian Classrooms

The Silent Indian Classroom Gap

- **One Fixed Speed:** Fast learners wait in boredom while struggling kids build on shaky Class 6–7 basics.
- **Compounding Gaps:** Chapter-end tests find gaps too late, right as Class 9–10 board pressure peaks.
- **Teachers Sidelined:** Coaching centers and passive video apps bypass the school teacher completely.



★ CORE PROBLEM SOLVED: Detects daily concept gaps in real time and restores the Indian school teacher to the center.

Steps 1–4: Cover Gateway, Dual Portals & 6-Digit Code

Steps 1 & 2: Cover & Dual Portals

- **Step 1 Orbital Cover:** Tapping the NEUROVA logo reveals core highlights, About Us, and contact footer.
- **Step 2 Role Split:** Dedicated Student Icon and Teacher Icon route users into specialized dashboards.
- **Zero Clutter UI:** Built for 11–to-16–year-old learners with instant 1–tap visual navigation.

Steps 3 & 4: Code + 4-Digit PIN

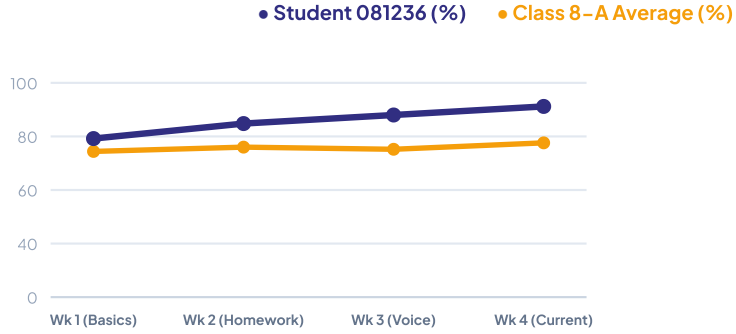
- **Step 3 Class Code:** Enters 6–digit code **081236**: 08 = Class 8, 1 = Sec A (2=B, 3=C), 36 = Roll No.
- **4-Digit Security PIN:** Sets a personal 4–digit PIN on 1st login & enters it next time for security.
- **Step 4 Profile Check:** Pre-loaded school roster fetches & confirms the student's exact profile.

★ KEY ADVANTAGE (STEPS 1–4): 6-digit Class Code (081236) + personal 4-digit PIN lets siblings securely share 1 family phone with zero tampering.

Steps 5–7: Student Vs. Class Average Line Graph

Where The Student Stands

- **Step 5 Standing View:** Immediately displays where confirmed Student 081236 stands across all topics.
- **Step 6 Real-Data Line:** Plots real student data from daily homework solving and oral voice explaining.
- **Step 7 Cohort Overlay:** Compares the student's trajectory line directly against the Class Average line.

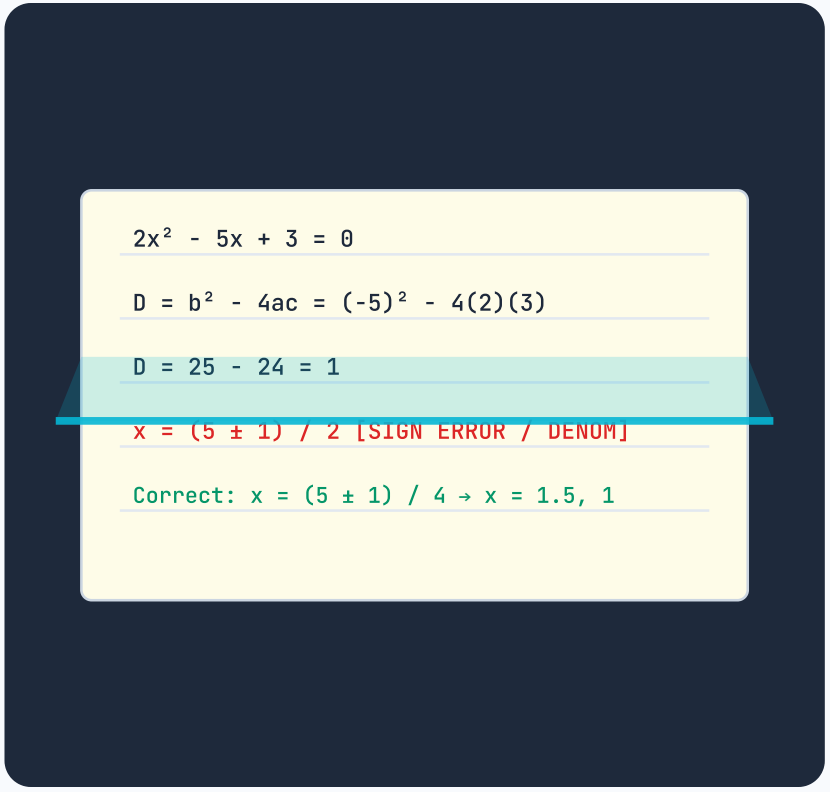


★ KEY ADVANTAGE (STEPS 5–7): Replaces blind end-of-term surprises with live, motivating percentile tracking against the class cohort.

Steps 8–9: AI Written & Text Analysis

Hyper-Granular Solution Auditing

- **Sir's 5–10 Daily Tasks:** Renders Sir's exact 5–10 daily homework questions for notebook solving.
- **Instant Blur & Ink Gate:** Sub-second OpenCV Laplacian scan flags blurry photos or messy ink to re-upload.
- **Deep Step-Level AI OCR:** Audits every intermediate equation line to catch formula vs. sign errors.

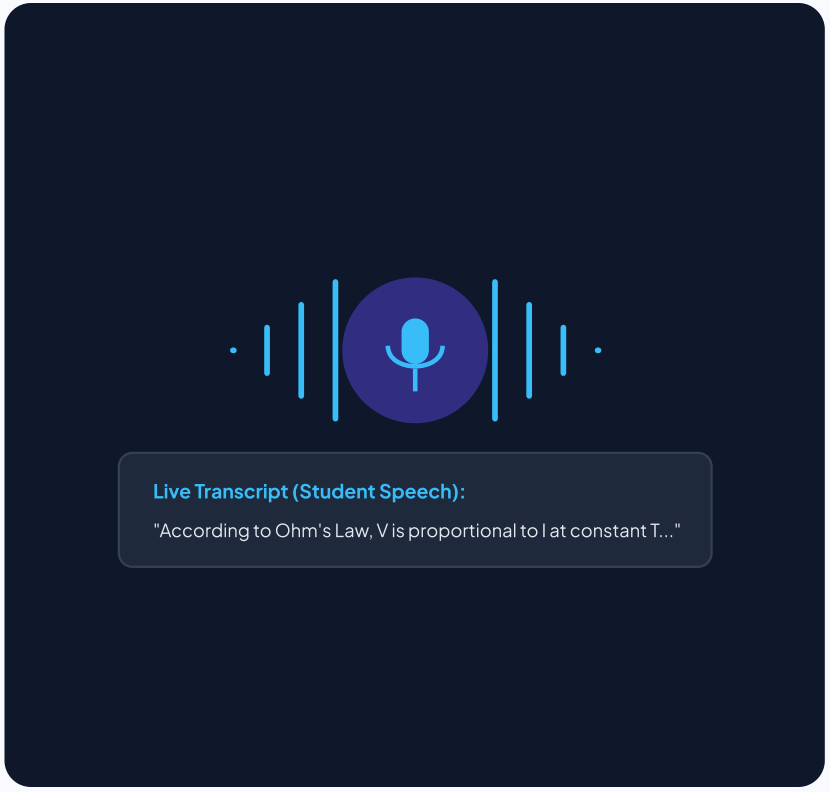


★ ADVANTAGE OF AI TEXT ANALYSIS: Finds the exact line where a student's logic broke down instead of just marking the final answer wrong.

Step 10: Voice AI & Live Transcript

Acoustic AI & Power Of Explaining

- **Neural Voice Model:** Bilingual acoustic AI analyzes vocal confidence, fluency, and semantic depth.
- **Editable Live Transcript:** Streams real-time text below the audio so students fix misheard words live.
- **Defeats Rote Copying:** A student can copy written homework, but cannot fake speaking a concept aloud.

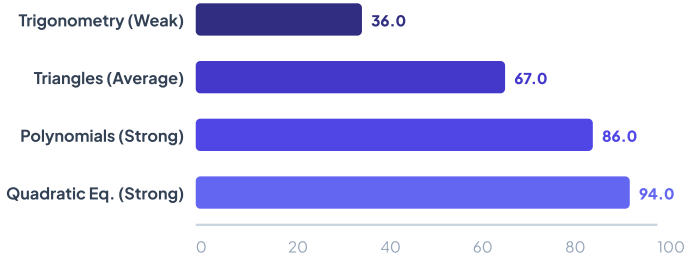


★ ADVANTAGE OF EXPLAINING ALOUD: Feynman oral articulation boosts memory retention by 3x and builds vernacular-to-English confidence.

Step 11: Cross-Modal AI Fusion (Strong, Average, Weak)

Written + Spoken AI Correlation

- **Dual-Engine Synthesis:** Fuses Step-9 handwritten text OCR with Step-10 semantic voice embeddings.
- **Detects Hidden Illusions:** Spots kids who write right steps by memory but fail oral concept reasoning.
- **Tri-Tier Visual Triage:** Classifies every sub-topic into 1) Strong, 2) Average, and 3) Weak graphs.



★ KEY ADVANTAGE (STEP 11): Eliminates false positives by verifying both procedural writing and conceptual speaking simultaneously.

Step 12: AI Voice Reteaching, Text Notes & 10 Flashcards

AI Voice Tutor

Voice Lesson: Explains Average & Weak topics one by one via voice.

Step Audio: Walks through missed steps in clear spoken words.

Text Deep-Dive

Text Notes: Follows voice audio with clear text explanations.

Key Formulas: Highlights exact theorems where the student got stuck.

10 Flashcards

10 Cards: Generates 10 flashcards for summary and revision.

Quick Recall: Locks key concepts into memory before re-testing.

★ KEY ADVANTAGE (STEP 12): Adapts to both auditory and visual learners instantly without needing expensive private coaching.

Step 13: Visual Scenarios & Voice Loop

Testing Beyond Rote Memorization

- **Real-Life Scenarios:** Generates real-world word problems using the concepts taught in Step 12.
- **Deep Visualization:** Requires mental 3D visualization of the situation instead of blind formula plugging.
- **Repeat Audio Loop:** Student must again explain the newly learned topic back to the AI by voice.



★ KEY ADVANTAGE (STEP 13): Builds real-world application skills required for competency-based CBSE/ICSE/State board questions.

Step 14: Final Mastery Unlock Or Teacher Flagging

If Solved

Test Passed: Solves the word problem and passes the voice check.

Show Progress: Proceeds ahead and updates final progress graphs.

If Still Weak

Gap Remains: Struggles on the problem even after AI re-teaching.

Escalate Now: Detects that the student needs teacher guidance.

Flag To Sir

Flag Student: Flags student and weak topic to Sir's dashboard.

Sir Re-Visits: Prompts Sir to personally re-visit that student.

★ KEY ADVANTAGE (STEP 14): Guarantees zero students slip through the cracks—combining AI self-remediation with human teacher care.

6 Hyper-Unique Innovations Of The NEUROVA Model

Cognitive & Multimodal Breakthroughs

- **1. Bimodal Truth Engine:** Only model fusing handwritten step OCR with live-editable spoken reasoning.
- **2. Sub-Second Blur Gate:** On-device OpenCV Laplacian filter rejects blurry notebook scans before upload.
- **3. 3D Visual Word Synth:** Replaces rote MCQs with real-life word problems requiring spatial visualization.

Classroom & Bharat-First Breakthroughs

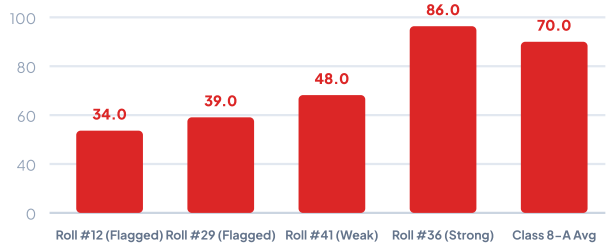
- **4. Code + 4-Digit PIN:** 6-digit code (081236) + 1st-time 4-digit PIN secures shared family phones.
- **5. Slip-Test Paper OCR:** Converts red-pen graded classroom paper tests into 100%/80% topic mastery bars.
- **6. Teacher Reteach Trigger:** Auto-alerts Sir to re-explain a topic when >50% of the section struggles.

★ COMPETITIVE MOAT: Every feature is purpose-engineered for Indian teachers, 50-student sections, and shared smartphones.

Teacher Console: Weak-Student Highlights & Reports

Exclusive Teacher Login & Views

- **Teacher Login + Code:** Sir selects the Teacher option, logs in securely, and enters the class code.
- **Weak-Student Graph:** Generates a class graph that specifically highlights weak students needing help.
- **1-on-1 Report & Flags:** Opens individual student diagnostic cards and lists all Step-14 flagged learners.



★ TEACHER ADVANTAGE: Gives Sir instant X-ray vision into a 50-student classroom without spending hours grading notebooks.

Empowering Teachers: Daily Telemetry & Reteach Alerts

Daily Automated Analysis To Sir

- **Evening Cohort Digest:** As students finish homework and voice tasks, AI sends daily analysis to Sir.
- **Complete Topic Split:** Delivers the full class breakdown of Weak, Average, and Strong topics.
- **Zero Grading Burden:** Eliminates manual checking while giving 100% visibility into student thinking.

Actionable Classroom Interventions

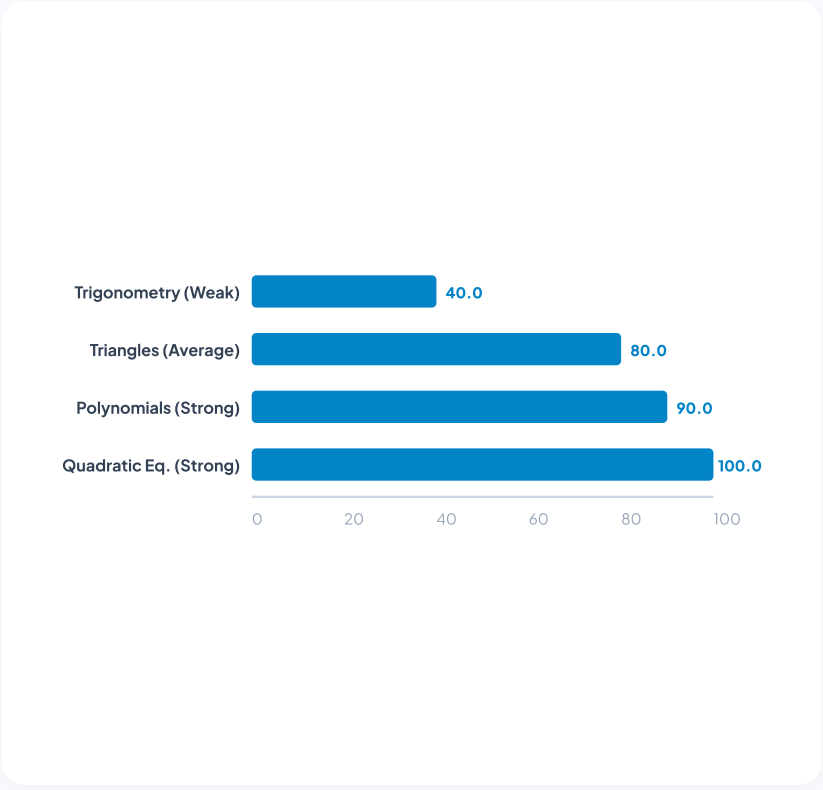
- **Majority Reteach Rule:** If a majority of the class fails a topic, Sir is alerted to re-explain it in class.
- **Multi-Topic Weak Focus:** Ranks students who are weak across multiple topics for 1-on-1 desk guidance.
- **Defeats Coaching Apps:** Empowers the school teacher to personalize teaching instead of sidelining them.

★ PROBLEM STATEMENT PILLAR #1: Directly fulfills the Decode Round mandate: puts the school teacher at the center of adaptive learning.

Classroom Slip-Test OCR And 100%/80% Mastery Bars

Paper Tests To Topic Bar Graphs

- **Question + Answer Upload:** Attaches class slip-test question papers and teacher-corrected answer sheets.
- **Granular Topic Mapping:** AI maps marks to topics: e.g., Strong in Quadratic Eq., Weak in Trigonometry.
- **100% / 80% Bar Structure:** Renders clear percentage bar graphs (100%, 80%, 40%) for student and teacher.



★ OFFLINE EXAM ADVANTAGE: Transforms ordinary red-pen paper slip-tests into granular topic mastery analytics in 10 seconds.

How Paper Slip-Tests Become AI Insights

Bridging Paper Exams & AI

- **In-Class Paper Tests:** Students write regular classroom unit tests and slip-tests on physical paper.
- **Teacher Red-Pen Grading:** Sir corrects the answer sheets and awards marks just like traditional schooling.
- **Dual Scan Analysis:** Uploading the question paper and graded sheet lets AI compute topic mastery bars.

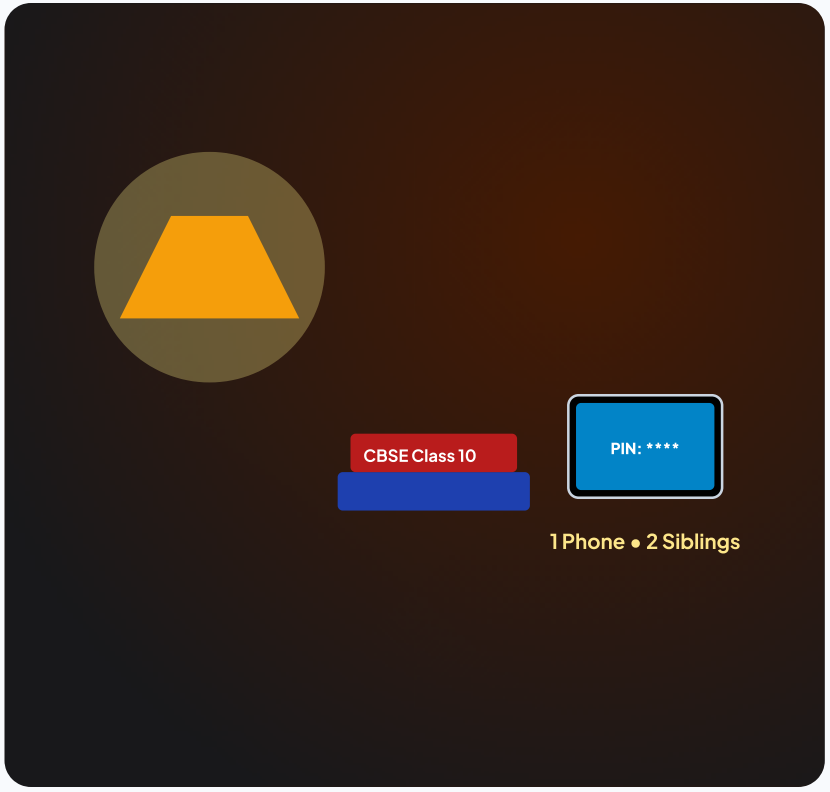


★ ZERO WORKFLOW DISRUPTION: Requires zero smartboards or school tablets—works with existing pen-and-paper Indian school exams.

Built For Real Indian Classrooms

Boards, Languages & Shared Phones

- **Shared Family Phone:** 6-digit code (081236) + 4-digit PIN lets siblings safely share one parent's phone.
- **Multi-Board & Vernacular:** Aligns with CBSE, ICSE & State Boards; bridges home language to school medium.
- **Low-Data & 50-Kid Rooms:** Runs on intermittent 2G/3G data and scales to 40-60 student Indian sections.



★ BHARAT-FIRST ADVANTAGE: Solves uneven device access, language transitions, and rising Class 6-to-10 board exam pressure.

Adapting Across Learn, Solve, Test In Indian Schools

Learn & Solve

Learn Phase: AI voice, text & 10 flashcards adapt to gaps.

Solve Phase: 5–10 daily tasks + blur flag + live transcript.

Test Phase

Word Problems: Real-life visual scenarios + voice re-test.

Slip-Test OCR: Scans class test papers into 100%/80% bars.

School Day

Home (15m): Scans homework, speaks summary & 10 flashcards.

Morning Class: Sir checks daily report & re-teaches weak topics.

★ DAILY HABIT ADVANTAGE: Fits into a 15-minute evening routine at home and a 2-minute morning teacher briefing at school.

17 · COMPREHENSIVE COMPARATIVE BENCHMARK

Traditional Approach Vs. NEUROVA Analysis System

Pedagogical Dimension	Traditional Classrooms & Global Apps	NEUROVA (Proposed Analysis System)
Student Onboarding	Complex email/password logins needing 1:1 devices	**6-digit code (081236) + 4-digit PIN** (created on 1st login)
Homework Verification	Manual checking or unverified photo uploads	**Instant AI blur/handwriting flag** + live editable voice transcript
Remediation (Learn)	One-pace lectures or generic video playlists	**Sequential AI Voice + Text notes + 10 auto-generated flashcards**
Testing Depth (Test)	Rote MCQ quizzes & delayed chapter-end exams	**Deep-visualization word problems + voice loop + slip-test OCR**
Teacher Empowerment	Teachers sidelined by coaching centers & AI apps	**Daily Weak/Average report**, majority reteach alert & student flags

★ **BENCHMARK ADVANTAGE: Outperforms both traditional single-speed classrooms and Western 1:1 laptop EdTech platforms.**

Which Technology Builds Which Feature? (Part 1)

NEUROVA Feature	Technology Used	How It Works
Cover & Login Page	HTML5 + CSS3	Creates and styles the cover logo halo, About Us, and login UI.
Student / Teacher Selection	HTML + JavaScript + CSS	Switches the selected portal role and updates the entry form.
6-Digit Code & 4-Digit PIN	JavaScript + Roster Store	Decodes 081236, verifies 4-digit PIN (set on 1st login) & profile.
Student & Teacher Dashboards	JavaScript + HTML + CSS	Displays student academic info and Sir's class statistics/actions.
Sections, Accounts & Tasks	JavaScript + HTML Forms	Creates Class 8A sections, accounts, and publishes 5-10 tasks.
Record Marks & Averages	JavaScript + HTML Tables	Validates assessment marks and calculates student/class averages.
Performance & Line/Bar Graphs	JavaScript + Canvas API	Draws student vs. class average line graphs & 100%/80% bars.

★ MODULAR WEB STACK ADVANTAGE: Zero heavy app-store downloads—runs instantly on any low-cost Android browser via HTML5, CSS3 & JS.

AI Vision, Voice & Data Tech (Part 2)

Feature	Tech Used	How It Works
Blur & Handwriting Gate	OpenCV.js + Vision AI	Flags blurry homework photos for instant re-upload.
Voice & Live Transcript	Web Speech + Whisper	Streams live editable text as student speaks.
AI Tutor & 10 Flashcards	LLM + Web Speech TTS	Explains weak topics by voice/text & builds 10 cards.
Visual Word & Slip-Test	Vision OCR + LLM	Generates word scenarios & scans graded slip-tests.
Teacher Alerts & Feedback	JS + HTML + Webhooks	Sends daily Weak/Avg reports & flags weak kids.
Offline Save & Responsive	localStorage + CSS3	Saves data across visits & fits shared family phones.

★ AI STACK ADVANTAGE: Combines edge OpenCV blur checks with cloud Vision OCR & Whisper ASR for 99% diagnostic precision.

Thank You! NEUROVA For Every Classroom.

Putting Indian School Teachers Back In Command While Adapting To Every Student.

NEUROVA · TEAM ANURAG, SIDDARTH, SOLMON, JITHIN (GRIET) | CODE KERNEL '26